██ Blue Ridge Stream Restoration & Mitigation LLC — Enterprise Repository

Welcome to the central B2B operations repository for **Blue Ridge Stream Restoration & Mitigation LLC**. This workspace is designed for Co-Founder & Managing Director **Hunter Morris** and Board Member **Hadi Irvani** to lead North Georgia's coldwater stream restoration and compensatory mitigation credit market.
This repository serves as a highly structured strategic, technical, and regulatory resource center, paired with a modern, responsive two-page web application designed for B2B credit buyers (hyperscale data centers, civil engineers, DOT planners) and private mountain landowners.

I. Core Philosophy: The Triple-Bank Semantic Framework

Traditional environmental conglomerates treat stream restoration as a sterile transaction, alienating rural families with complex regulatory jargon. Blue Ridge Stream Restoration overcomes this friction by leveraging the triple-meaning of the word **Bank** to unify ecology, regulation, and family wealth into a single storytelling pillar:

```
                            [THE TRIPLE-BANK FRAMEWORK]

       1. THE GEOMORPHIC BANK        2. THE REGULATORY BANK        3. THE WEALTH & LAND BA
NK
     (Fluvial & Spawning Ecology)    (Savannah District Ledger)      (Landowner Family Leg
acy)
              |                               |                               |
    Vertical bank regrading          Savannah District IRT          70/30 Joint Venture sp
lit
     to a stable 3:1 slope           accreditation, credit           yielding $696k payout
s or
    with Rhododendron cover          deposits on RIBITS database   zero-cost streamside up
grades
```

1. **The Geomorphic Bank (Physical Stream Bed)**: We repair vertical, eroding pasture banks, regrading them to stable 3:1 slopes and transplanting mature *Rhododendron maximum* to create thermal canopy shade, keeping water temperatures below 65°F for native trout spawning.
2. **The Regulatory Bank (Savannah District Credit Ledger)**: We register the geomorphic improvements under USACE Savannah District guidelines, converting ecological restoration into certified, high-yield stream mitigation credits deposited in the federal RIBITS database.
3. **The Wealth & Land Bank (Landowner Financial Legacy)**: We sell these credits to exurban commercial developers or hyperscale data center campuses (QTS, Microsoft, Google) at premium pricing. Under our 70/30 Joint Venture model, landowners receive a **30% cash payout split** (e.g. $696,600 on our 1,500-LF Anderson Creek reach) or premium civil streamside upgrades constructed by Hunter's active 12-to-20+ field crew at zero out-of-pocket cost.

II. Repository Directory Structure (PhD-Grade Reorganization)

The repository has been programmatically sorted into 10 highly organized, color-coded subdirectories, aligning all strategic, technical, and legal assets into structured tiers:

```
ACR Stream Restoration/
■■■ 01_Strategic_&_Business_Plans/       # Vision documents, recruiting plans, Traction
EOS scorecards
■■■ 02_Regulatory_&_USACE_Frameworks/    # USACE banking rules, legal guidelines, bioeng
ineering manuals
■■■ 03_Project_Case_Studies/             # Flagship geomorphic case studies and NWP 27 P
CN templates
■■■ 04_Market_&_Competitor_Intelligence/  # Savannah District registries and exurban data
 center leads
■■■ 05_Sales_&_Outreach_Playbooks/       # Warm B2B outreach scripts, email structures,
and playbooks
■■■ 06_Legal_&_Cooperative_Agreements/    # eSignature playbooks and land mitigation deed
 covenants
■■■ 07_Marketing_&_Social_Content/       # LinkedIn strategies, brand style guides, and
cinematic videos
■■■ 08_Recruiting_&_Internships/         # 12 specialized UGA/Clemson/GT intern training
 workstreams
■■■ 09_Operations_&_Technical_Playbooks/  # Drone photogrammetry, RTK GNSS, and hosting s
etups
■■■ 10_Outreach_&_Relationship_Banking/   # Ag/timber banker databases and B2B referrals
directories
■■■ index.html                           # Public Landowner Siting Page & Yield Calculat
or
■■■ portal.html                          # Private Operations & Strategic Resource Porta
l
■■■ styles.css                           # Global Trout-HSL Design Tokens & Layout style
sheet
```

III. Web Application Architecture

The public-facing portal is constructed using a modern, performant **Vanilla HTML5, CSS3, and ES6 JavaScript** architecture, bypassing heavy Javascript frameworks for lightning-fast page loading and 100% SEO optimization.

1. Public Landowner Sourcing Page (`index.html`)

Designed to capture exurban B2B credit requests and source high-yield rural stream tracts in North Georgia:

- **Trout-Stewardship Contrast**: Highlights Hunter Morris's niche as North Georgia's leading outfitter (directing a crew of 12 to 20+ active river professionals), contrasting geomorphic stream craftsmanship against corporate conglomerates that clear-cut and use heavy concrete rip-rap.
- **Tabs Case Studies Console**: Interactive, responsive CSS-tabs showcasing commercial stream credits (Roya's Cabin/Anderson Creek) vs trout habitat step-pool restoration (Hunter's Cabin/Goldmine Hollow).
- **Free Stream Restoration Yield Estimator**: Real-time Javascript calculator allowing landowners to enter their stream length and bank erosion severity to project stream credit yields, covered CapEx, and landowner gross cash payouts under the 70/30 split.

2. Private Operations & Strategic Resource Portal (`portal.html`)

The secure, B2B central database for partners, lawyers, and relationship lenders:

- **Glassmorphic Folder Selector**: Responsive dashboard equipped with HSL trout-coordinated folder cards, custom vector SVG operational icons, and active indicator animations.
- **Dynamic Real-Time JavaScript Search & Filters**: A comprehensive, programmatic search algorithm that filters and renders all **55 enterprise files** in real time by operation tiers, subdirectories, titles, and descriptions.
- **High-Definition Widescreen Video Players**: Widescreen video players (1920x1080, 16:9 ratio) featuring slow-motion leaps of wild trout and slow-motion casting zooms, narrated in a deep, warm nature documentary style (David Attenborough tone) to establish high-end brand authority.

3. Styling Core (`styles.css`)

Establishes the design tokens and variables used across all web layouts:

- **HSL Palette Tokens**: River Navy `hsl(209, 59%, 29%)`, Trout Teal `hsl(168, 54%, 37%)`, Copper Rust/Accent `hsl(18, 63%, 43%)`, and Warm Gold `hsl(43, 80%, 54%)`.
- **Responsive Flexbox/Grid**: Clean container grids with smooth transitions and glassmorphic micro-animations.

IV. Dynamic Calculations & Engineering Rigor

All geomorphic and financial calculators running in the background utilize verified fluvial geomorphology and Savannah District SOP equations:

1. Geomorphic Soil-Loss Equation

Calculates the exact tonnage of sediment prevented from choking local trout spawning gravels annually:

```
Sediment Prevented (Tons/Year) = (L * H * Re * ρb) / 2000
```

Where:

- `L` = Stream reach length (Linear Feet)
- `H` = Stream bank height (Feet)
- `Re` = Lateral erosion rate (Feet/Year), measured using bank pins.
- `ρb` = Soil bulk density constant (`85.0 lb/ft³` for North Georgia sand-clay matrices).
- `2000` = Conversion constant from pounds to short tons.

2. Boundary Shear Stress Equation

Ensures instream logs, root-wads, and J-hook structures remain rock-solid during 100-year storm events:

```
Boundary Shear Stress (τ) = γ * R * S
```

Where:

- `τ` = Fluvial Boundary Shear Stress (lb/ft^2 or N/m^2)
- `γ` = Unit weight of water (62.4 lb/ft^3 or 9,810 N/m^3)
- `R` = Hydraulic Radius (ft or m), calculated as cross-sectional bankfull area (A) divided by wetted perimeter (P).
- `S` = Channel slope gradient (ft/ft or m/m).

V. Quick GitHub Pages Deployment Guide

Hunter Morris can easily deploy and share this active B2B portal in under 2 minutes utilizing **GitHub Pages** for $0.00 in hosting costs:

Step 1: Create a GitHub Repository

1. Log into GitHub at github.com.
2. Click **New** to create a new repository.
3. Name the repository: `ACR-Stream-Restoration`.
4. Choose **Public** or **Private** (GitHub Pages supports private repositories on Pro plans; select Public for free hosting).
5. Leave "Add a README" unchecked (we already have this master README).

Step 2: Push Local Files to GitHub

Open your terminal inside the local directory `/Users/irvani/Library/CloudStorage/GoogleDrive-hadi.irvani@gmail.com/My Drive/ACR Stream Restoration` and execute:

```
# Initialize git (if not already done)
git init

# Add the remote GitHub origin (replace <your-username> with your actual GitHub username)
git remote add origin https://github.com/<your-username>/ACR-Stream-Restoration.git

# Set main branch
git branch -M main

# Push the committed files to GitHub
git push -u origin main
```

Step 3: Enable GitHub Pages

1. On your GitHub repository page, click on **Settings** (gear icon on the top menu).
2. On the left sidebar, under the "Code and automation" section, click on **Pages**.
3. Under the **Build and deployment** section, select **Deploy from a branch** as the Source.
4. Under the **Branch** dropdown, select **main** and set the folder to `/ (root)`.
5. Click **Save**.

Step 4: Access and Share the Portal!

GitHub will generate a live, secure URL for your venture within 60 seconds. The URL will follow this structure:
■ `https://<your-username>.github.io/ACR-Stream-Restoration/`
You can immediately copy and send this link to:

- **Hunter Morris** to view the dynamic dashboard.
- **Private Landowners** to estimate their geomorphic credit split payouts.
- **B2B Buyers & Developers** to check active credit inventories and download CWA Section 404 compliance guidelines.

VI. Team & Academic Leveraged Staffing model

Blue Ridge Stream Restoration operates at hyper-scale speeds using an **Academic-Leveraged model**: achieving a 10-person capacity firm with only 3 Full-Time Equivalents (FTEs). We rotate specialized graduate interns to run our core technical pipelines:

- **University of Georgia (UGA) Warnell School of Forestry**: Runs field biology, EPA rapid bioassessments, HOBO water probes, and vegetative buffer surveying.
- **Georgia Institute of Technology**: Formulates 1D/2D HEC-RAS hydraulic models, processes RTK survey thalwegs, and drafts AutoCAD Civil 3D bioengineering construction plans.
- **Clemson University MRED Program**: Handles real estate NPV underwriting, records conservation easements, and audits escrow balances.

Developed by Hunter Morris & Hadi Irvani. Built with geomorphic and operational integrity.